

WRRRA Spatial Twist Selection of Galactic Disk Planes and the Dark Mass Phenotype

A Dual Output Model of the Milky Way Warp and Rotation Curve

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Research status Quantitative toy model and falsifiable WRRRA hypothesis

Abstract

This paper formulates the selection of an approximately two-dimensional galactic disk plane in three-dimensional space as a problem of WRRRA spatial twist. The central revision is to separate the roles of twist and angular momentum. Twist neither creates rotation nor determines its direction. The directional sector of twist selects a local plane field that does not distinguish up from down, while its magnitude sector is stored as spatial stress and generates additional gravity that appears as dark mass. Angular momentum is a separate physical quantity that subsequently maintains rotation within the selected plane and prevents central collapse.

This distinction is applied to the Milky Way. A thin-disk approximation to the McMillan baryonic mass model is combined with the fixed WRRRA transition scale $a_t = 1.1918 \times 10^{-10} \text{ m s}^{-2}$ and a single constitutive response calibrated with SPARC. Without galaxy-specific recalibration, the resulting rotation speed differs from a linearized version of the Eilers et al. 5-25 kpc rotation curve by an RMS of 2.42 km s^{-1} over 8-25 kpc. The model gives 226.3 km s^{-1} at the solar radius of 8.2 kpc and 201.6 km s^{-1} at 25 kpc. The disk plane is calibrated to the onset of the Gaia DR3 Cepheid warp at $R \approx 11 \text{ kpc}$, an outer inclination of about 3 degrees, and the radial change of the line of nodes. These results do not prove the twist hypothesis. They provide the first closed model in which one twist state produces disk shape and additional gravity as distinct outputs.

Keywords WRRRA spatial twist galactic disk Milky Way warp dark matter mass phenotype rotation curve

Core Assessment

Stage	Content	Current assessment
Validation inputs	Milky Way baryonic model and rotation curve Gaia warp H_0 f_t	External observations
WRRRA transformation	Unoriented plane field $[n]$ and projector P constitutive twist-stress response v	Calculated in this paper
Outputs	Disk plane z_t g_t v WRRRA equivalent twist mass	Quantitative outputs
Falsification condition	The same twist state fails to pass warp rotation vertical equilibrium and lensing jointly	Untested
Conclusion	The first Milky Way toy model works numerically but the causal hypothesis remains unconfirmed	Conditional pass

1 Research Question

The existence of angular momentum in a galactic disk does not by itself answer why matter gathered into that plane. Standard gas cooling and angular-momentum conservation are powerful mechanisms for flattening and maintaining a disk. WRRRA asks a prior geometrical question: how is a particular two-dimensional transport plane selected in three-dimensional space, and why does that plane bend and twist with position?

This paper proposes that spatial twist first selects an unoriented plane, after which matter and angular momentum align with and operate within that plane. Here unoriented means that up and down are not assigned. A plane has two opposite normals, but both represent the same plane. A local disk plane can therefore be defined without introducing a privileged global vertical direction.

The second question is whether another sector of the same twist state participates in dark matter phenomena. The existing WRRRA twist-stress model separates the direct gravity of ordinary matter g_m from the additional gravity of spatial stress g_t [1,2]. This study places the scalar magnitude sector and the directional sector that selects the disk plane within one twist state, without forcing them to be identical or governed by the same equation.

2 Elements Preserved from the Existing WRRRA Model

This paper does not alter WRRRA Core or Minimal Computation Cosmology 2.3.2. Standard physical descriptions of ordinary matter and the four interactions are retained. Instead of immediately declaring a fifth force, additional gravity is calculated in the weak-field Renderer as an effective source in which twist stress stored in space is expressed as mass. The source branch of general relativity and the modified-response branch are not counted twice in the same calculation.

Earlier work normalized the present-universe twist stress with $H_0=67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $f_t=0.265$, giving $a_t=1.1918 \times 10^{-10} \text{ m s}^{-2}$. Across the 201 lowest-acceleration SPARC points, this value differed from the diagnostic inverse estimate by about 0.75 percent. The present paper transfers that normalization to the Milky Way without refitting it.

The new element is a computational layer for disk-plane selection. A scalar χ_t or v that represents only the magnitude of twist stress cannot select a plane. The minimal state must therefore record a plane director that has direction but no sign, together with energy density and anisotropic stress.

3 Separating the Roles of Twist and Angular Momentum

3.1 Unoriented Plane

Let $\mathbf{n}$ be a unit normal representing a local plane. Because $\mathbf{n}$ and $-\mathbf{n}$ represent the same plane, the physical state is the equivalence class $[\mathbf{n}]$, not the vector itself. The plane projector P is invariant under this sign reversal.

$$\mathbf{n} \sim -\mathbf{n}, \quad P_{ij} = \delta_{ij} - n_i n_j, \quad P^2 = P$$

This structure does not specify up and down in advance. It also does not specify clockwise or counterclockwise rotation. The twist plane field therefore neither defines angular momentum nor hides the sign of angular momentum within its definition.

3.2 Shape of the Surface

The mean disk plane is represented as a level set $S=0$. With this definition, position-dependent normals construct one continuous surface.

$$S(R, \phi, z, t) = z - z_T(R, \phi, t) = 0, \quad \mathbf{n} = \frac{\nabla S}{\|\nabla S\|}$$

For a general normal field to integrate into a physical surface, the Frobenius integrability condition must hold. Because this paper constructs the normal from S , the condition is satisfied by construction.

$$\mathbf{n} \cdot (\nabla \times \mathbf{n}) = 0$$

3.3 Magnitude Sector of the Same Twist State

To separate the directional sector of twist from its gravitational magnitude sector, the effective stress contains an energy density ρ_t , isotropic pressure p_t , and anisotropic stress π_t defined relative to the unoriented normal. The sign and magnitude of q_t represent the difference between in-plane and out-of-plane responses.

$$T_T^{\mu\nu} = \rho_T u^\mu u^\nu + p_T h^{\mu\nu} + \pi_T^{\mu\nu}, \quad \pi_{T,ij} = q_T \left(n_i n_j - \frac{1}{3} \delta_{ij} \right)$$

The quantity ρ_t controls the magnitude of additional gravity, whereas $[n]$ and q_t control the orientation of the disk plane and its vertical restoring response. The sectors belong to one state but are not the same number. This is the minimal decomposition that turns the statement twist selects the plane but does not define angular momentum into a calculable model.

4 Constitutive Equation for the Gravitational Sector

4.1 Matter Gravity and Twist Stress Gravity

For weak-field circular orbits, the observed gravitational acceleration is decomposed directly into a baryonic component and a spatial-stress component.

$$g_{\text{obs}}(R) = g_M(R) + g_T(R) = \nu\left(\frac{g_M}{a_T}\right) g_M(R)$$

An empirical response that is consistent with the full SPARC range is adopted and frozen as the calibrated constitutive equation of this model. This is a model revision that promotes a relation previously used as external support into the actual Milky Way calculation. No claim is made that the functional form has been derived from a WRR action.

$$\nu(y) = \frac{1}{1 - \exp(-\sqrt{y})}, \quad g_T = [\nu(y) - 1] g_M$$

The transition scale is fixed by the cosmological normalization.

$$a_T = c H_0 \sqrt{\frac{f_T}{8}} = 1.1918 \times 10^{-10} \text{ m s}^{-2}$$

4.2 Source Formulation and Prohibition of Double Counting

This paper uses the weak-field branch in which $\rho_{t,\text{eff}}$ is read as an additional effective source. The same effect is not added again as a modified-geometry term. A covariant completion would require conservation of the total stress-energy tensor.

$$\nabla^2 \Phi = 4\pi G(\rho_b + \rho_T^{\text{eff}}), \quad \nabla_\mu(T_b^{\mu\nu} + T_T^{\mu\nu}) = 0$$

The quantity $\rho_{\text{t,eff}}$ is not a particle number density. It is the mass phenotype obtained when the same additional gravitational field is inverted as a Newtonian mass source. The equivalent twist mass in this paper must therefore not be read as a measurement of actual particle mass.

5 Milky Way Validation Inputs

The baryonic inputs are taken from the McMillan mass model of the Milky Way [3]. They include the central surface densities and scale lengths of the thin and thick stellar disks, the masses and scale lengths of HI and H₂ gas, and the bulge mass. The original model includes finite thickness and central gas holes, whereas the present rotation calculation approximates each component as a thin exponential disk. This simplification cannot describe the bar and bulge region inside 5 kpc precisely.

Component	Mass $10^9 M_\odot$	Scale kpc	Representation
Thin stellar disk	35.66	2.53	Exponential disk
Thick stellar disk	11.25	3.38	Exponential disk
HI gas	11.00	7.00	Exponential disk
H ₂ gas	1.20	1.50	Exponential disk
Bulge	9.13	a=0.70	Hernquist approximation

The rotation-curve comparison uses the 5-25 kpc result from the analysis of more than 23,000 red giant stars by Eilers et al. [4]. That study reported 229.0 km s⁻¹ at the solar radius and an outer slope of -1.7 km s⁻¹ kpc⁻¹. The present paper uses a linearized target built from those two reported values rather than the full individual data set. The RMS values below are therefore first-order shape comparisons, not observational likelihood tests.

Warp inputs are taken from Gaia DR3 Cepheid studies. Dehnen et al. reported no clear warp inside about 11 kpc, an inclination that increases outward to about 3 degrees, and a line of nodes that changes with radius [5]. Cabrera-Gadea et al. reported m=1 and m=2 components, a twisted line of nodes for R>11 kpc, and an m=1 pattern speed of about 9.2±3.1 km s⁻¹ kpc⁻¹ [6]. Low-order m=0,1,2 modes also dominate the HI disk [7].

6 Calculation Method

6.1 Baryonic Rotation Field

The circular speed of each exponential disk is calculated with the standard thin-disk expression containing modified Bessel functions.

$$v_d^2(R) = 4\pi G \Sigma_0 R_d y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)], \quad y = \frac{R}{2R_d}$$

The bulge is represented by a Hernquist approximation with mass M^b and scale a^b .

$$v_b^2(R) = \frac{GM_b R}{(R + a_b)^2}$$

The squared velocities of the components are added to obtain $g_m = v_{\text{bar}}^2/R$. The total circular speed is then calculated using the fixed v and a_i .

$$v_{\text{WRRRA}}(R) = \sqrt{R g_{\text{obs}}(R)}$$

6.2 Twist Plane and Warp

The observed mean disk height can be expressed in low-order Fourier modes. This is the standard form used in existing warp analyses [5-7]. WRRRA interprets it not merely as the posterior shape of a material disk but as the observational representation of the twist plane field $S=0$.

$$z_T = Z_0(R) + Z_1(R)\sin[\phi - \phi_1(R, t)] + Z_2(R)\sin(2[\phi - \phi_2(R, t)])$$

The quantitative demonstration retains only the fundamental $m=1$ mode. The warp begins at 11 kpc, reaches 3 degrees at 14 kpc, and then saturates. The line of nodes changes by about 14 degrees per kpc over the observed interval and is conservatively held fixed beyond 16.5 kpc.

$$Z_1(R) = R \tan i(R), \quad i(R) = 3^\circ \text{ clip}\left(\frac{R-11}{3}, 0, 1\right), \quad \phi_1(R) = -15^\circ + 14^\circ \text{ clip}(R - 11, 0, 5.5)$$

Excluding the $m=0$ and $m=2$ modes does not deny their existence. The purpose is first to show how much can be reconstructed with one directional field, while leaving the lopsided warp as a structured residual for the next validation stage.

7 Outputs

7.1 Rotation Curve

R kpc	Baryonic v	g/g_m	WRRRA v	Target v	$M_{t,\text{eff}} 10^{10} M_\odot$
5.0	192.4	0.319	221.0	234.4	1.37
8.2	182.6	0.537	226.3	229.0	3.41
12.0	161.0	0.868	220.1	222.5	6.28
16.5	138.9	1.320	211.6	214.9	9.77
20.0	125.9	1.692	206.6	208.9	12.48
25.0	112.0	2.237	201.6	200.4	16.32

Over 8-25 kpc, the RMS difference between the WRRRA curve and the linearized Eilers target is 2.42 km s^{-1} , the mean difference is -2.14 km s^{-1} , and the maximum absolute difference is 3.31 km s^{-1} . Over the full

5-25 kpc range, the RMS is 3.64 km s^{-1} . The largest discrepancy occurs near the inner boundary, where the bar and non-axisymmetry are strongest.

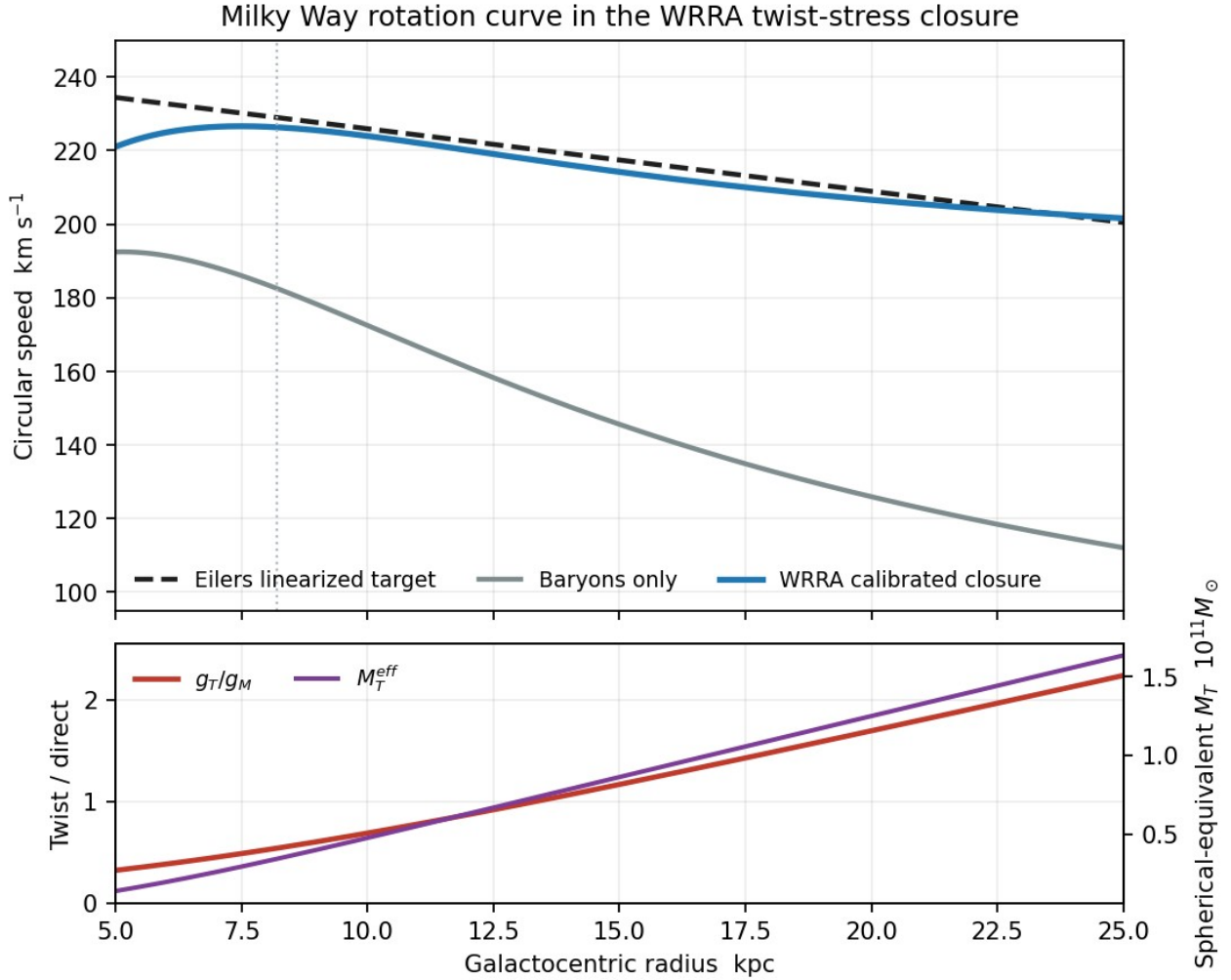


Figure 1 Milky Way rotation curve calculated from the McMillan baryonic inputs and the fixed WRRa constitutive response. The dashed line is the linearized observational target from Eilers et al.

Neither a_t nor the functional form of v was readjusted to fit the Milky Way rotation curve. The baryonic model itself is based on Milky Way observations, however, and v is a SPARC-calibrated constitutive relation, so this is not a model-independent blind prediction. Under the WRRa assessment rule, it is classified as a Milky Way output obtained by freezing validated inputs and applying the same transformation.

7.2 Equivalent Twist Mass

The additional acceleration can be converted into a spherically equivalent Newtonian mass as follows.

$$M_T^{\text{eff}}(< R) = \frac{R^2 g_T(R)}{G}$$

This equivalent mass is $3.41 \times 10^{10} M_\odot$ at 8.2 kpc, $1.25 \times 10^{11} M_\odot$ at 20 kpc, and $1.63 \times 10^{11} M_\odot$ at 25 kpc. If the physical twist stress has a two-dimensional or anisotropic distribution, its energy-density profile is not

identical to this spherical equivalent mass. The quantity is a ledger of the additional gravity required by the rotation field.

7.3 Disk Plane and Warp Shape

R kpc	Inclination deg	Line of nodes deg	m=1 maximum height kpc
5.0	0.00	Undefined	0.000
8.2	0.00	Undefined	0.000
12.0	1.00	-1.0	0.209
16.5	3.00	62.0	0.865
20.0	3.00	62.0	1.048
25.0	3.00	62.0	1.310

Minimal m=1 Milky Way warp used as the twist-plane calibration

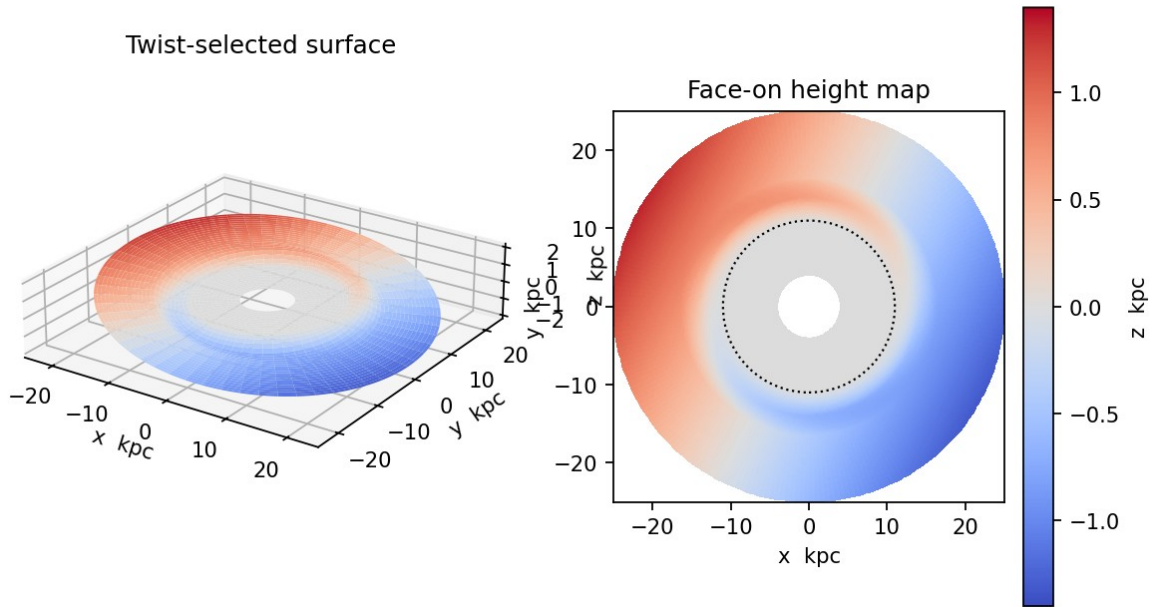


Figure 2 Minimal m=1 realization of the WRR twist plane using the Gaia Cepheid warp as a calibration input The dotted circle marks the 11 kpc warp onset radius

This plane was not constructed by entering an angular-momentum vector. The observed warp shape is used to fix the twist plane field, after which the degree to which stellar and gas orbits align with that plane remains a separate dynamical problem. The current figure shows only the mean plane and does not include disk thickness, m=2 asymmetry, spiral arms, or the bar.

7.4 Two Outputs from One Twist State

The minimal twist state in this paper is the tuple consisting of an unoriented plane $[n]$, stress energy density ρ_t , and anisotropic magnitude q_t . It produces the disk plane, additional gravity, and vertical restoring force as distinct Renderer outputs.

$$\mathcal{T} = ([n], \rho_T, q_T) \implies \{z_T(R, \phi, t), g_T(R), \Phi_{\perp}\}$$

Testing vertical motion in the disk requires a restoring potential for the distance S away from the plane. At lowest order it may be written as follows.

$$\Phi_{\perp} \simeq \frac{1}{2} v_z^2 S^2, \quad h_z \simeq \frac{\sigma_z}{v_z}$$

The model is complete only when v_z can be calculated from ρ_t and q_t fixed by the rotation curve. At present, no constitutive equation for q_t has been specified, so h_z is not predicted numerically. It is important not to hide this gap.

8 Distinctive WRRR Explanation and the Role of Existing Physics

Question	Standard physical explanation	Connection added by WRRR
Why is the disk flat	Gas collisions and radiative cooling reduce vertical motion	The twist plane supplies the reference plane relative to which vertical residual motion is removed
Why that plane	A plane perpendicular to the total angular-momentum vector	An unoriented transport plane $[n]$ is hypothesized to precede angular momentum
Why does the outer disk warp	Satellite interactions and halo torques	A position-dependent plane field z_t and anisotropic stress produce the warp
Why is there additional gravity	A particle dark matter halo or modified gravity	The mass phenotype g_t and $\rho_{t,eff}$ of spatial stress

WRRR does not remove angular-momentum conservation, gas cooling, or the collisionless motion of stars. Its originality does not lie in inventing already known numerical values. It lies in separating and connecting plane selection and the dark-mass phenotype as directional and magnitude sectors of one twist state. Even when another theory uses the same rotation curve or warp equation, outputs calculated through the WRRR state and transformation remain WRRR outputs.

Conversely, merely citing an external warp equation or RAR constitutive relation without calculating it inside WRRR would not count as integration. This paper includes both relations as calibrated model components and calculates Milky Way outputs, so they are integrated into the numerical model. Deriving those constitutive relations from a deeper WRRR action remains an open task.

9 Falsification Conditions and Observation Plan

The hypothesis is not validated merely because a disk is observed. Standard cooling and tidal torques can also produce disks and warps. The discriminatory power of WRRRA comes from joint constraints left by the same twist state across different observables.

- If g_t and $p_{t,eff}$ fixed by the rotation curve cannot reproduce the stellar vertical velocity dispersion and disk thickness, the directional-stress coupling fails.
- If the twist stress required by dynamics cannot reproduce weak and strong gravitational lensing through the same metric, the dark-mass phenotype interpretation fails.
- If the warp plane z_t cannot connect the mean planes and vertical velocity fields of stars, HI, and H_2 through one time evolution, the plane-field interpretation fails.
- If the $m=2$ lopsidedness and line-of-nodes variation are fully explained by instantaneous torques from external satellites, with no residual left for a fixed spatial-stress directional field, the upper bound on the additional WRRRA term approaches zero.
- If polar-ring galaxies or mutually perpendicular multiple disks cannot be represented by one local plane field and arbitrary extra planes must be introduced, the minimal-computation hypothesis is weakened.
- If a_t or v must be freely refitted for each galaxy, the universal constitutive equation for spatial stress is falsified.

The most direct next calculation is to solve the vertical Jeans equation within the same Milky Way baryonic model and constrain q_t . After fixing g_t from the rotation curve, the model must simultaneously fit the solar-neighborhood K_z , outer-disk flaring, Cepheid V_z , and HI thickness. This test determines whether the disk plane and dark gravity are truly two outputs of the same twist state.

10 Limitations

- The $m=1$ warp shape is a calibrated input fitted to Gaia observations. It was not derived from first principles.
- The empirical SPARC relation v is integrated into and calculated within this model, but it has not been uniquely derived from a twist action.
- The baryonic model simplifies the McMillan model into thin axisymmetric disks. It omits central gas holes, the bar, spiral arms, and non-circular motion.
- The Eilers target is a curve linearized from two reported values. It is not a likelihood test using the individual data points and their covariance.
- The equivalent twist mass is a spherical inverse estimate and is not the physical mass distribution of a two-dimensional stress energy field.
- The constitutive equation for q_t and the vertical restoring frequency v_z have not yet been fixed, so the model does not predict disk thickness.
- A covariant completion that simultaneously calculates structure growth, the CMB, cluster collisions, and lensing lies outside the present scope.

11 Conclusion

This paper presents a calculable WRRRA hypothesis that differs from the explanation that angular momentum creates the disk plane. The directional sector of spatial twist first selects a plane that does not distinguish up from down, while angular momentum maintains rotation within that plane. An initially global vertical direction or a two-dimensional rotation plane therefore need not be supplied in order for an approximately two-dimensional disk to arise in three-dimensional space.

The magnitude sector of twist produces additional gravity g_t under the existing WRRRA model and appears as the dark-mass phenotype. Using the Milky Way baryonic model with fixed a_t and fixed v , the calculated curve differs from the linearized observed rotation curve by an RMS of 2.42 km s^{-1} over 8-25 kpc. The

directional sector implements the observed warp beyond 11 kpc and an inclination of about 3 degrees as one continuous plane.

The present result does not identify the disk plane and dark gravity with one equation. Instead, it separates and calculates them as two outputs from one twist state. The final assessment is that the first closure of the quantitative Milky Way toy model succeeds, while the claim that spatial twist is the physical cause of the disk plane remains unconfirmed. The next falsification stage is to determine whether the stress field fixed by the rotation curve also passes vertical-equilibrium and lensing tests without additional calibration.

Appendix A Reproducible Values

Item	Value	Status
H_0	$67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Validation input
f_t	0.265	Validation input
a_t	$1.191812669 \times 10^{-10} \text{ m s}^{-2}$	WRRR closure output
Stellar disk mass	$46.910 \times 10^9 \text{ M}_\odot$	Converted McMillan input
Gas mass	$12.2 \times 10^9 \text{ M}_\odot$	McMillan input
Bulge mass	$9.13 \times 10^9 \text{ M}_\odot$	McMillan input
WRRR vc at 8.2 kpc	226.34 km s^{-1}	Output
WRRR vc at 25 kpc	201.59 km s^{-1}	Output
M_{eff} at 25 kpc	$1.632 \times 10^{11} \text{ M}_\odot$	Spherical equivalent output
Warp onset	11 kpc	Observational calibration input
Outer inclination	3°	Observational calibration input

Appendix B Claim Grades

Claim	Grade	Basis for assessment
An unoriented plane is defined without up or down	Mathematical fact	P is invariant under $n \rightarrow -n$
The twist plane precedes the material disk plane	WRRH hypothesis	Requires cross-validation with vertical equilibrium
Twist stress generates additional gravity	WRRH causal hypothesis	First rotation test passed lensing and growth untested
Reproduction of the Milky Way rotation curve	Post-calibration output	Calculated from fixed a_t and v with baryonic inputs
Reconstruction of the Milky Way warp surface	Observationally calibrated model	Gaia inclination and line-of-nodes variation used as inputs
Common cause of the two outputs	Integration hypothesis	q_t and v_z remain unspecified

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